If $2+\sqrt{3}$ is a root of a quadric polynomial equation with rational coefficients, then $2-\sqrt{3}$ is also a root; thus their product $(x-(2+\sqrt{3}))(x-(2-\sqrt{3}))$ is a factor; that is x^2-4x+1 is a factor; we can divide the polynomial with this factor and get the quotient as a second degree factor which can be solved by known techniques. Using this, we can find all the roots of the quadric equation. This technique is applicable for all surds taken in place of $2+\sqrt{3}$.

If an imaginary root and a surd root of a sixth degree polynomial with rational coefficient are known, then step by step we may reduce the problem of solving the sixth degree polynomial equation into a problem of solving a quadratic equation.

Example 3.15

If $2+i$ and $3-\sqrt{2}$ are roots of the equation

$$x^6-13x^5+62x^4-126x^3+65x^2+127x-140=0,$$

find all roots.

Solution

Since the coefficient of the equations are all rational numbers, and $2+i$ and $3-\sqrt{2}$ are roots, we get $2-i$ and $3+\sqrt{2}$ are also roots of the given equation. Thus $(x-(2+i))$, $(x-(2-i))$, $(x-(3-\sqrt{2}))$ and $(x-(3+\sqrt{2}))$ are factors. Thus their product

$$((x-(2+i))(x-(2-i))(x-(3-\sqrt{2}))(x-(3+\sqrt{2}))$$

is a factor of the given polynomial equation. That is,

$$(x^2-4x+5)(x^2-6x+7)$$

is a factor. Dividing the given polynomial equation by this factor, we get the other factor as (x^2-3x-4) which implies that 4 and -1 are the other two roots. Thus

$$2+i, 2-i, 3+\sqrt{2}, 3-\sqrt{2}, -1, \text{ and } 4$$

are the roots of the given polynomial equation.

3.7.2 Polynomial equations with Even Powers Only

If $P(x)$ is a polynomial equation of degree $2n$, having only even powers of x, (that is, coefficients of odd powers are 0) then by replacing x^2 by y, we get a polynomial equation with degree n in y; let $y_1, y_2, \cdots y_n$ be the roots of this polynomial equation. Then considering the n equations $x^2=y_r$, we can find two values for x for each y_r; these $2n$ numbers are the roots of the given polynomial equation in x.

Example 3.16

Solve the equation $x^4-9x^2+20=0$.

Solution

The given equation is

$$x^4-9x^2+20=0.$$

This is a fourth degree equation. If we replace x^2 by y, then we get the quadratic equation

$$y^2-9y+20=0.$$

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